Knowledge Graph Management Systems

Vadalog

Advanced

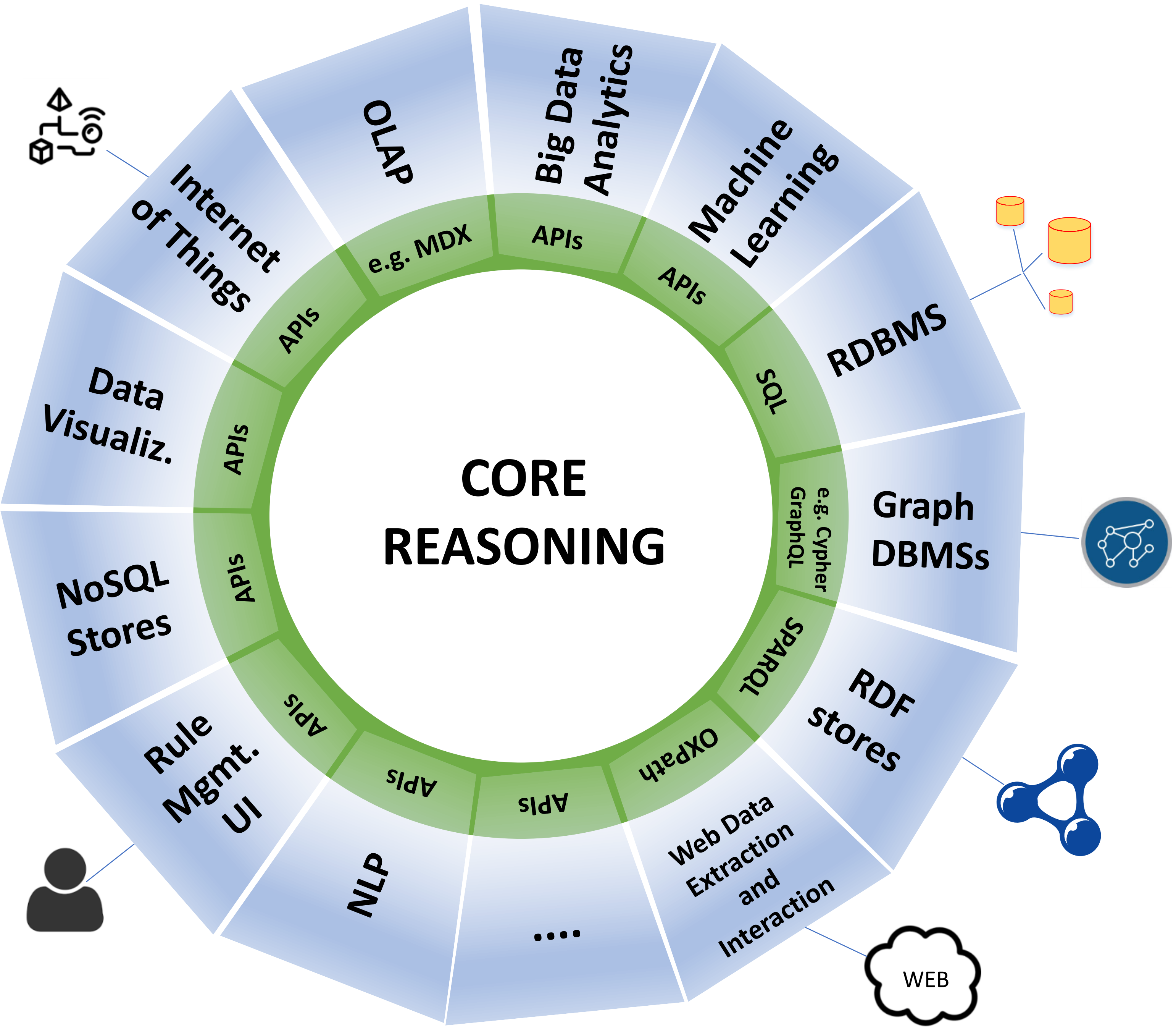
Emanuel Sallinger

WARNING: Advanced Unit

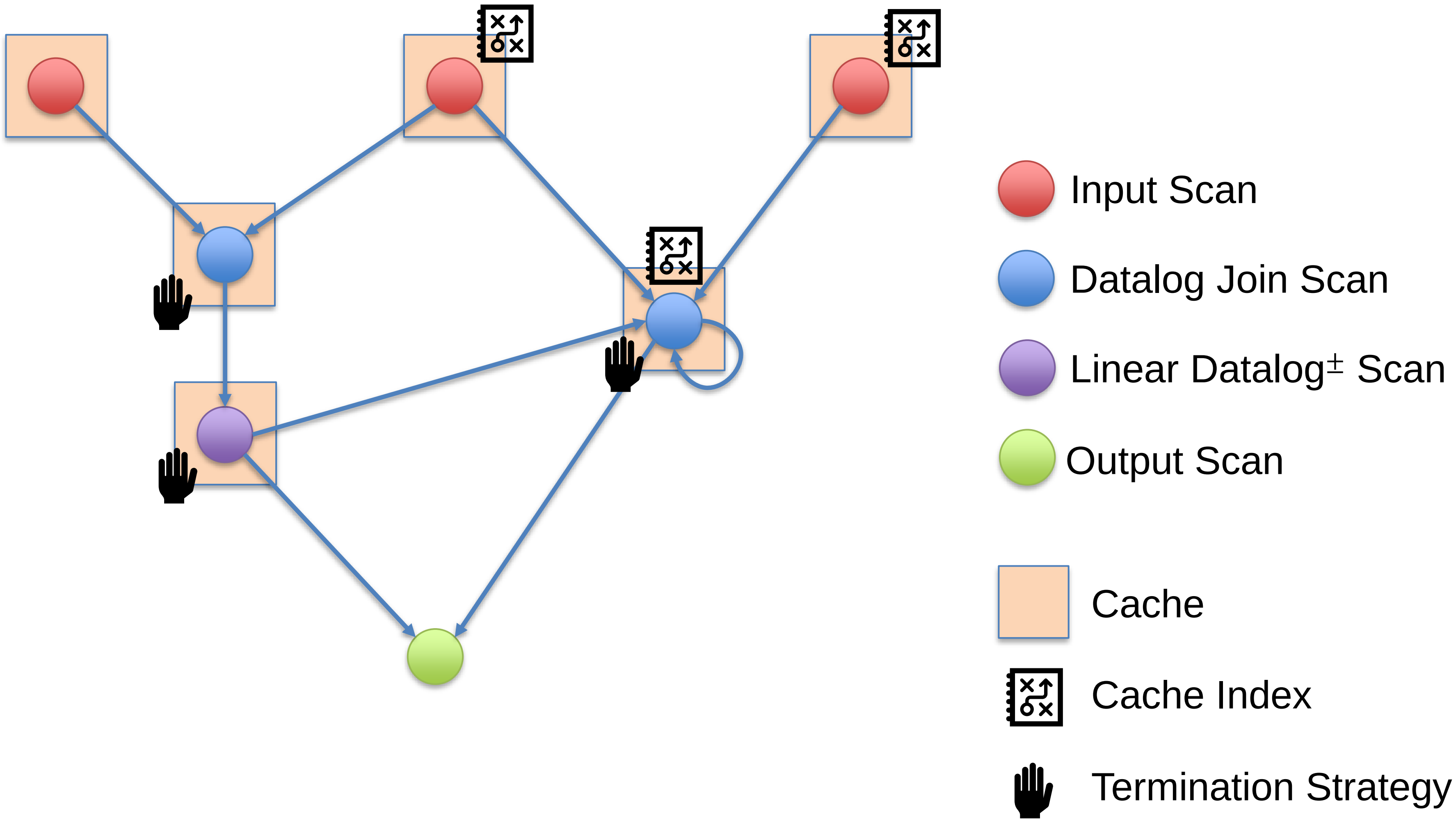
Only for the interested

Needs further reading to understand all concepts

Bellomarini, Sallinger, Gottlob.
The Vadalog System: Datalog-based Reasoning for Knowledge Graphs.
VLDB, 2018.


CORE
REASONING
OLAP
e.g. MDX
Big Data
Analytics
APIs
Machine
Learning
APIs
RDBMS
SQL
Graph
DBMSs
e.g. Cypher
GraphQL
RDF
stores
SPARQL
Web Data
Extraction
and
Interaction
OXPath
WEB
....
APIs
NLP
APIs
Rule
Mgmt.
UI
APIs
NoSQL
Stores
APIs
Data
Visualiz.
APIs
Internet
of Things
APIs

In-Memory Stream Processing



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\begin{aligned}
1 &: \text{Company}(x) \to \exists p \exists s \; \text{Owns}(\hat{p}, \hat{s}, x)\\
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3 &: \text{Owns}(\hat{p}, \hat{s}, x) \to \text{PSC}(x, \hat{p})\\
4 &: \text{PSC}(x, \hat{p}), \text{Controls}(x, y) \to \exists s \; \text{Owns}(\hat{p}, \hat{s}, y)\\
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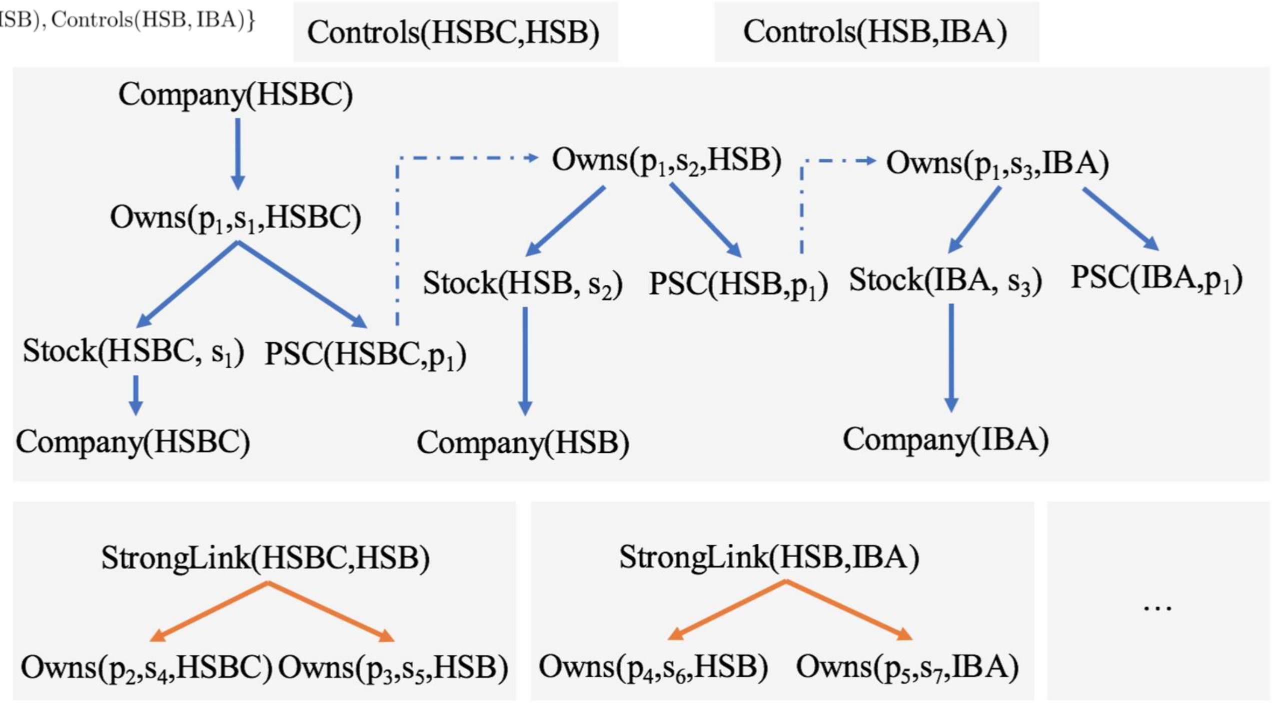

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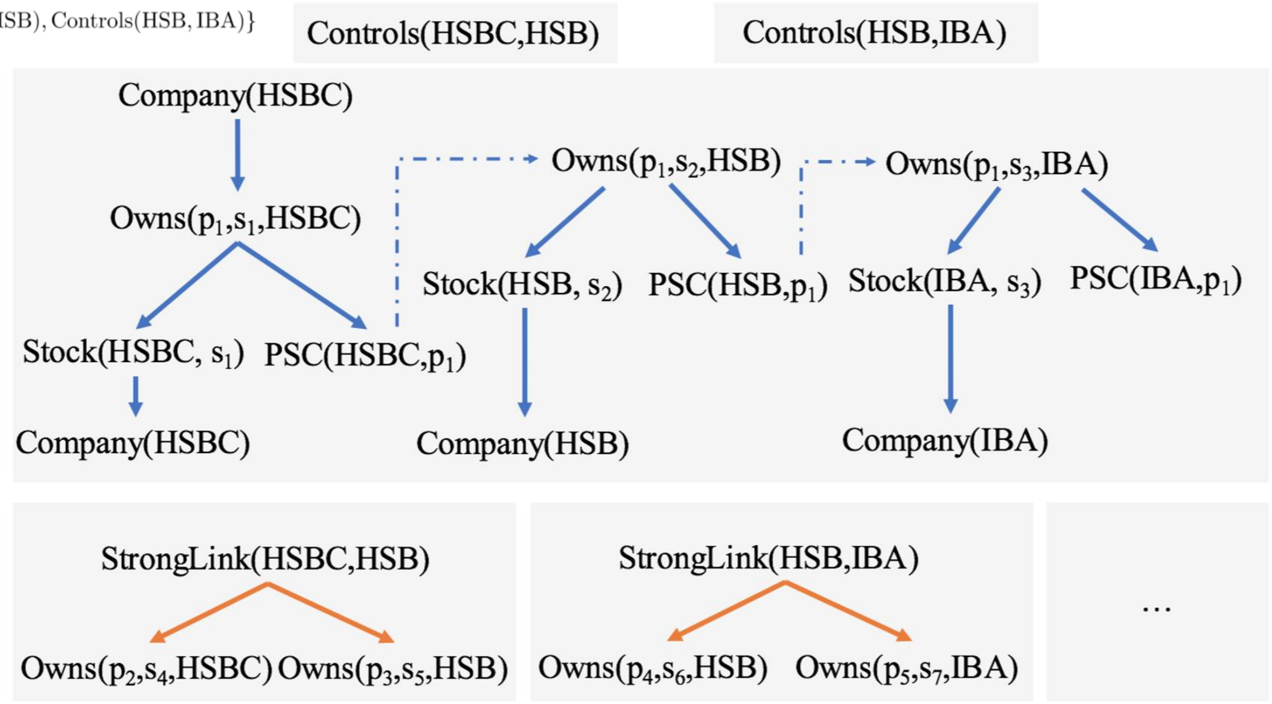

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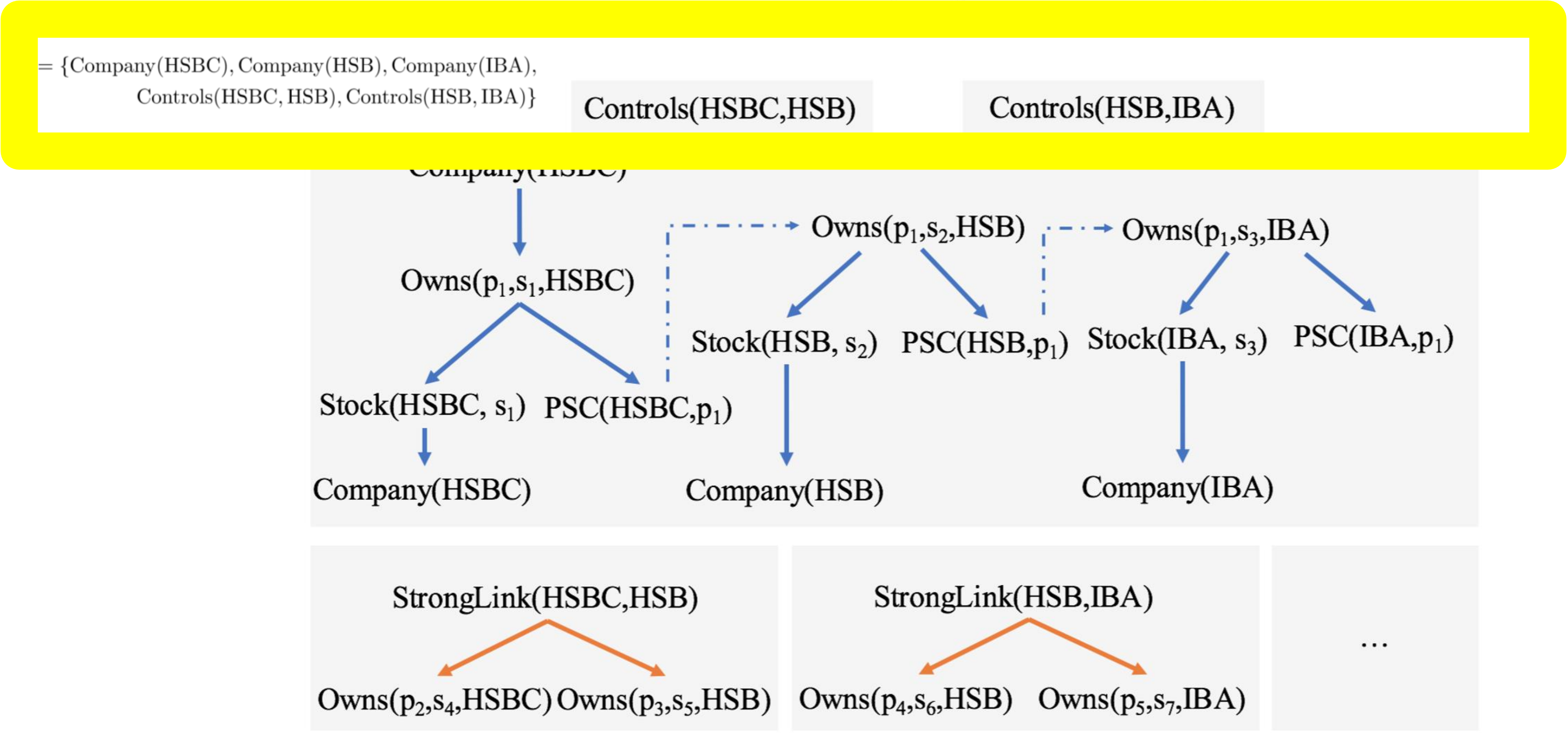

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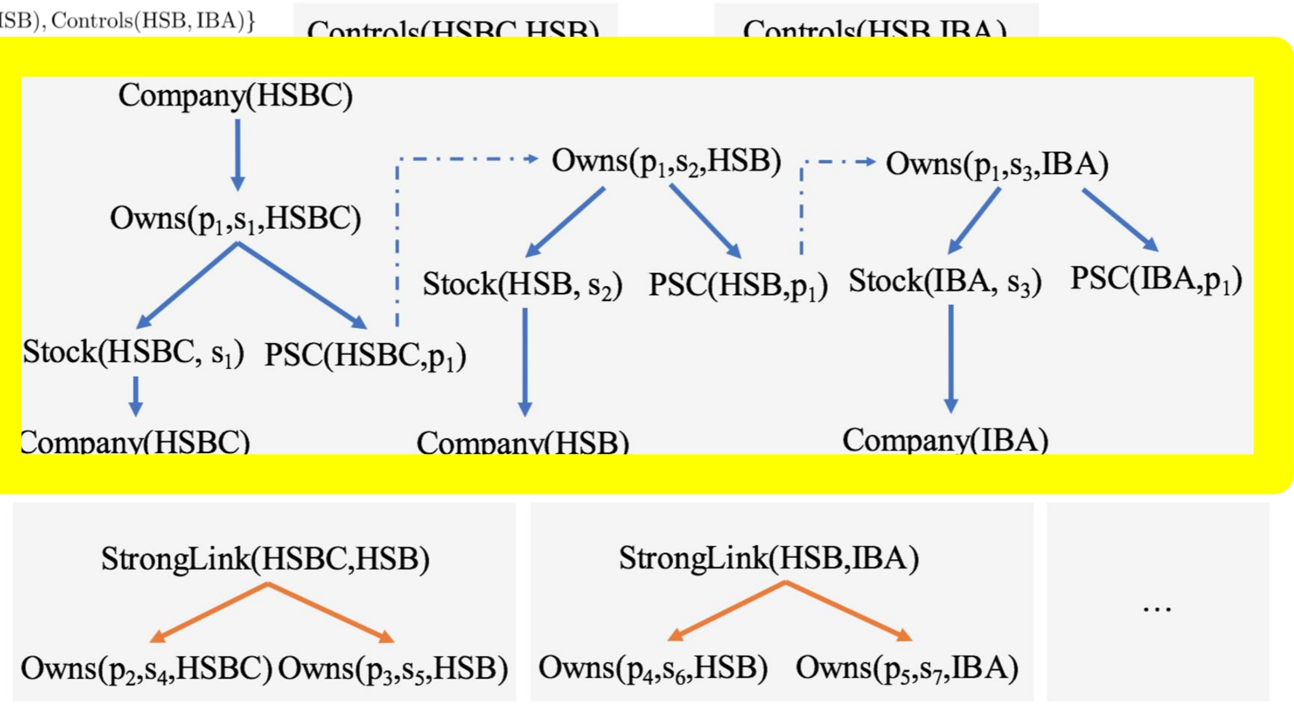

Dangerous

$$\mathrm{Emp}(x) \rightarrow \exists \boldsymbol{z}\, \mathrm{Mgr}(x, \boldsymbol{z}) \qquad \underbrace{\mathrm{Mgr}(x, \boldsymbol{y})}_{\text{Ward}} \wedge \mathrm{Pers}(x) \rightarrow \mathrm{Emp}(\boldsymbol{y})$$

Ward

1. all the "**dangerous**" variables should coexist in a single body-atom α, called the **ward**

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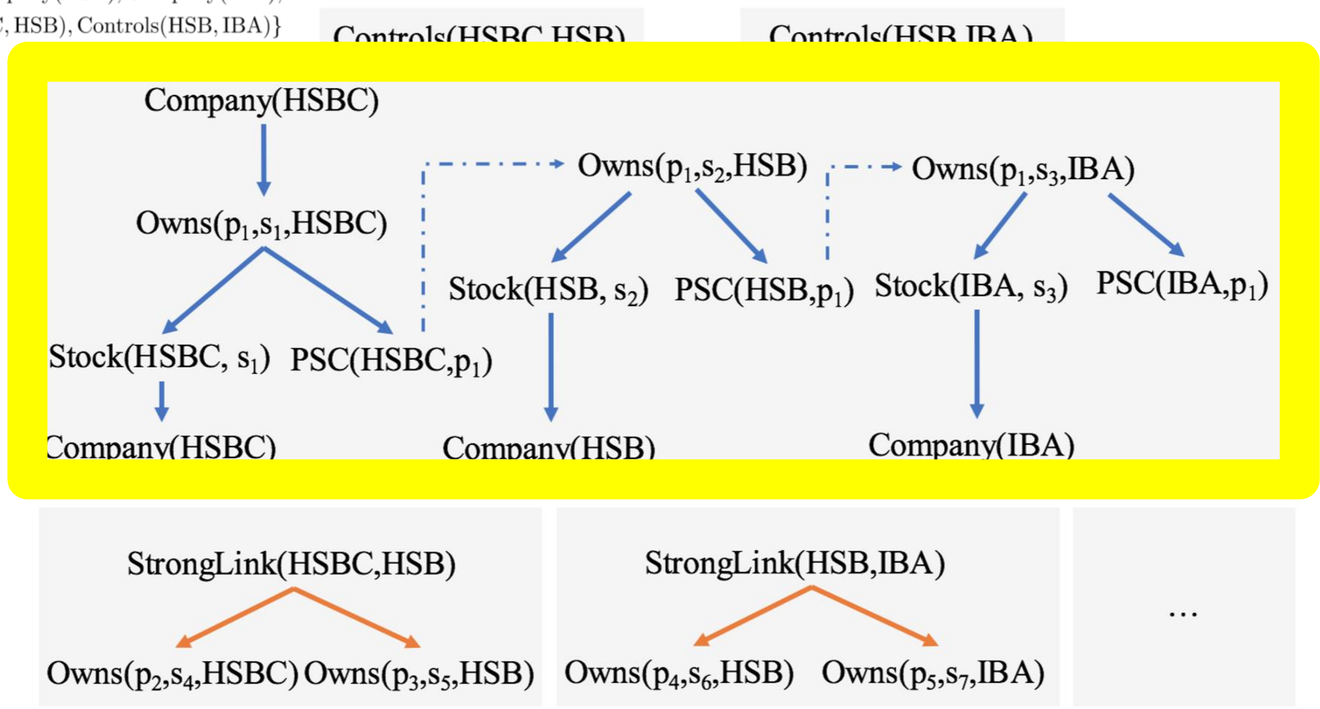
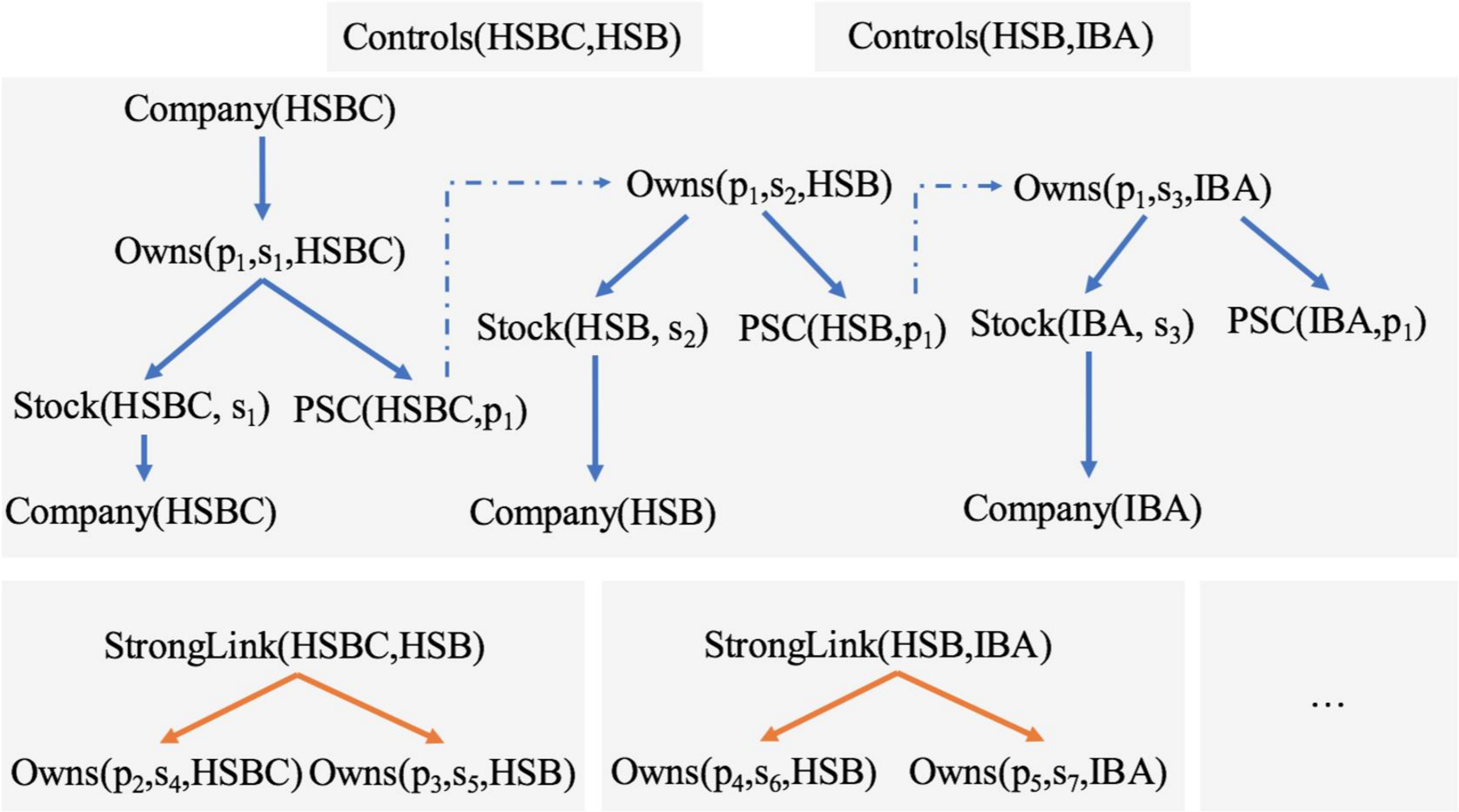


Theorem

Let a and b be two facts in a warded forest. If they are isomorphic, then $subtree(a)$ is isomorphic to $subtree(b)$.



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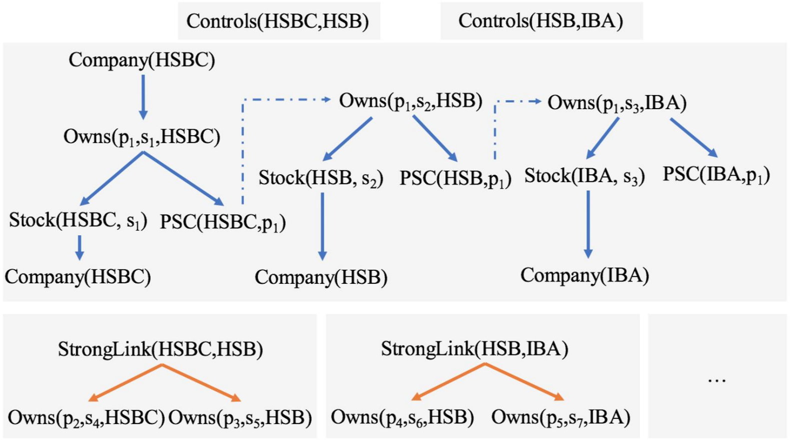
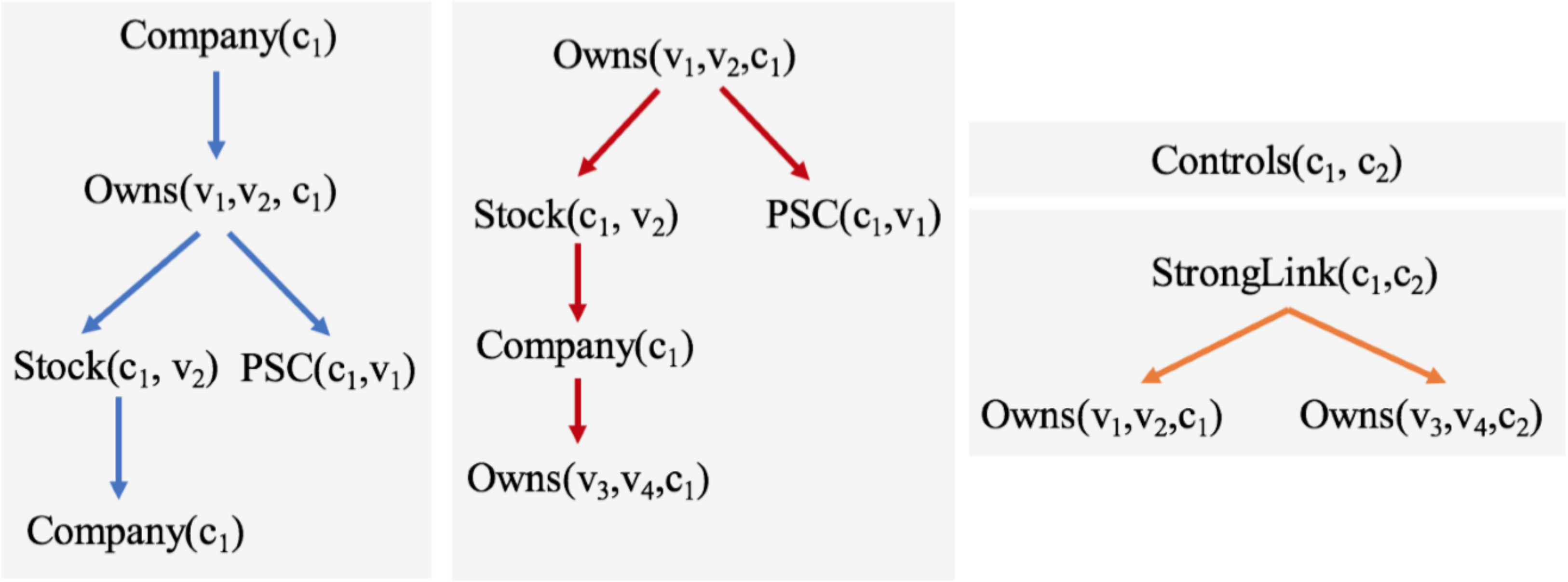


Theorem

Let a and b be two facts in the chase graph of a set of harmless warded rules. If a and b are isomorphic, then $subgraph(a)$ is isomorphic to $subgraph(b)$.

Proposition

Let a and b be two facts in a linear forest. If they are pattern-isomorphic, then $subtree(a)$ is pattern-isomorphic to $subtree(b)$.



Algorithm 1 Termination strategy for the chase step.

function CHECK_TERMINATION(**a**)
  if a.generating_rule == {LINEAR or WARDED} **then**
    if $\exists\lambda \in$ S[π(**a**.l_root)] s.t. $\lambda \subseteq$ **a**.provenance **then**
      return false ▷ beyond a stop provenance
    else if $\exists\lambda \in$ S[π(**a**.l_root)] s.t. **a**.provenance $\subset \lambda$ **then**
      return true ▷ within a stop provenance
    else ▷ continue exploration
      if $\exists$**g** *in* G[**a**.w_root] s.t. **a** isomorphic to **g** **then**
        S[π(**a**.l_root)]=**a**.provenance
        return false ▷ isomorphism found
      else
        G[**a**.w_root].append(**a**)
        return true ▷ isomorphism not found
  else if a $\notin$ G **then** ▷ other non-linear generating rules
    G[**a**.w_root].append(**a**) ▷ and reset provenance
    return true
  else ▷ the new tree is redundant
    return false

Algorithm 2 A generic chase using the termination strategy.

function CHASE(D, Σ)
  for all $\sigma \in \Sigma$ and **x** to which σ applies **do**
    if CHECK_TERMINATION($\sigma(\mathbf{x})$) **then**
      $D = D \cup \{\sigma(\mathbf{x})\}$

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